Understanding Linear Discriminant Analysis (LDA)

1. LDA may perform poorly when:
    - A) Classes are well separated
    - B) Data satisfies Gaussian assumptions
    - C) Classes have significant nonlinear boundaries
    - D) Features are informative
2. Which assumption of LDA is often violated in real-world datasets?
    - A) Equal covariance matrices across classes
    - B) Presence of labels
    - C) Numerical features
    - D) Finite samples
3. LDA can struggle when:
    - A) The number of samples is much larger than the number of features
    - B) The number of features exceeds the number of samples
    - C) There are only two classes
    - D) Features are standardized
4. Why can LDA be sensitive to outliers?
    - A) It uses class means and covariance estimates
    - B) It performs clustering
    - C) It ignores feature values
    - D) It uses decision trees
5. Which of the following is a limitation of LDA?
    - A) Requires labeled data
    - B) Assumes normally distributed classes
    - C) Assumes equal class covariances
    - D) All of the above
6. When class covariance matrices are substantially different, which method may outperform LDA?
    - A) Quadratic Discriminant Analysis (QDA)
    - B) PCA

- C) K-Means
- D) KNN Regression

7. LDA assumes that class boundaries are:
   - A) Linear
   - B) Circular
   - C) Arbitrary
   - D) Fractal
8. What problem arises when the within-class scatter matrix S_W is singular?
   - A) Projection becomes easier
   - B) Matrix inversion becomes difficult or impossible
   - C) Classes become balanced
   - D) Dimensionality increases automatically
9. LDA may not effectively separate classes when:
   - A) The classes are linearly separable
   - B) The classes overlap heavily
   - C) Features are relevant
   - D) Labels are correct
10. In high-dimensional datasets with few samples, LDA often suffers from:
    - A) Overfitting and unstable covariance estimates
    - B) Underfitting only
    - C) Reduced computation time
    - D) Automatic feature selection
11. Which data characteristic can negatively impact LDA performance?
    - A) Strong outliers
    - B) Noisy measurements
    - C) Non-Gaussian distributions
    - D) All of the above
12. LDA is generally less suitable than kernel-based methods when:
    - A) Data is linearly separable
    - B) Data contains nonlinear class structures
    - C) Data has few features
    - D) There are only two classes

13. A major limitation of LDA compared to PCA is that:
    - A) It requires class labels
    - B) It can handle more dimensions
    - C) It is unsupervised
    - D) It ignores covariance
14. Why might class imbalance affect LDA?
    - A) Larger classes can dominate mean and covariance estimation
    - B) LDA automatically balances classes
    - C) LDA ignores sample counts
    - D) Class imbalance improves separability
15. For a dataset with C classes, a limitation of LDA is that the maximum number of discriminant components is:
    - A) Equal to the number of samples
    - B) Equal to the number of features
    - C) $C - 1$
    - D) Unlimited
16. Which statement best describes a limitation of classical LDA?
    - A) It assumes a common covariance structure across all classes
    - B) It works perfectly for any nonlinear dataset
    - C) It does not require labels
    - D) It is immune to outliers
17. LDA may produce suboptimal results when:
    - A) Feature distributions deviate significantly from normality
    - B) Classes are clearly separated
    - C) Data is low-dimensional
    - D) Covariances are equal
18. Which of the following is known as the "small sample size problem" in LDA?
    - A) Too many classes compared to features
    - B) Too many features compared to samples
    - C) Too many samples compared to features
    - D) Too few labels
19. LDA cannot extract more than $C - 1$ discriminant features because:

- A) The rank of the between-class scatter matrix is limited
- B) NumPy restricts dimensions
- C) Eigenvalues are always zero
- D) PCA removes extra dimensions

20. Which scenario is most likely to violate LDA assumptions?
    - A) Two Gaussian classes with equal covariance matrices
    - B) Multiple classes with highly nonlinear boundaries and unequal covariances
    - C) Balanced classes with low noise
    - D) Well-separated classes in low dimensions